As Predicted:"*(December 2019) Self-Punishing for Fortuitous Reward – Study 2*" (#31335)

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Author(s)
How Hwee Ong (Tilburg University) - h.h.ong@tilburguniversity.edu
Rob Nelissen (Tilburg University) - Rob.Nelissen@tilburguniversity.edu
Ilja van Beest (Tilburg University) - i.vanbeest@uvt.nl

1) Have any data been collected for this study already?
No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?
Prior research (Kramer, Irmak, Block, & Ilyuk, 2012; Wright, da Costa Hernandez, Sundar, Dinsmore, & Kardes, 2013) and our recent work suggests that people expect suffering to bring about reward even when no such causal link exists. Building on this finding, we hypothesize that choosing to self-inflict suffering would increase the perceived likelihood of subsequently experiencing a fortuitous outcome.

3) Describe the key dependent variable(s) specifying how they will be measured.
At the start of the study, participants are informed that they will be entered into a lottery where they have the chance to win a financial reward. The outcome of this lottery will be made known after the completion of the study.

The key dependent variable is the perceived likelihood of winning the lottery (hereafter 'likelihood'). This will be measured on visual analogue scale ranging from "No chance of winning" to "Definitely will win". This dependent variable will be measured at three time points over the course of the study (more details later on in this section).

Another variable is participants' choice between two versions of a task that purportedly measures 'finger agility' (hereafter 'version choice'). One of the two versions will ostensibly cause mild and temporary hand pain while the other does not involve any mention of pain. Choosing the painful version is indicative of the tendency to self-inflict suffering. This measure is adapted from a prior study by Slepian and Bastian (2017). After choosing the version, participants will subsequently perform the 'finger agility task'.

The order in which the dependent variables will be measured is as follows. After participants read about the lottery, 'likelihood' will be measured for the first time (Time #1). Next, participants will indicate their 'version choice'. Participants will then indicate 'likelihood' (Time #2). Participants will proceed to perform the finger agility task. After which, 'likelihood' will be measured for the third time (Time #3).

4) How many and which conditions will participants be assigned to?
The study uses within-subject design in which we measure 'likelihood' at three time points.
Participants' 'version choice' serves as a non-manipulated between-subject grouping variable.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.
The main analysis examines our prediction that choosing the painful version of the finger agility task would increase the perceived likelihood of winning the lottery to a larger extent than choosing the task version without pain. To this end, we will conduct a 2 (Time: Time #1 vs. Time #2) x 2 (Version Choice: Pain vs. No Pain) mixed ANOVA. We predict a significant interaction effect, which would then be followed up with planned contrasts that separately examine the change in likelihood between Time #1 and Time #2 for each of the two between-subject groups (Pain vs. No Pain).

For exploratory purposes, we will conduct the same set of 2 x 2 mixed ANOVA mentioned above except that Time #2 will be replaced by Time #3. This explores if the effect would persist even after participants completed the finger agility task.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.
After performing the 'finger agility task', participants will be asked to recall the version of the task they had previously chosen. Only participants who provide a correct response will be included in analyses.

7) How many observations will be collected or what will determine sample size?
No need to justify decision, but be precise about exactly how the number will be determined.
The study will be conducted during a single class lecture. Sample size will be solely determined by the number of students who volunteer to participate in the study during the lecture session.

We project that about 500 students will volunteer to participate in the study. Assuming an exclusion rate of 10% would leave us with 450 participants. Sensitivity power analysis conducted using G*Power indicate that with a sample size of 450 and correlation between repeated measures at r = .5, we will have 80% power to detect an interaction effect of f = .066 in the mixed ANOVA specified in the analysis section. We deemed this level of power to be adequate.

8) Anything else you would like to pre-register?
(e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)
For exploratory purposes, we also ask the following questions:
o Self-reported score on the finger agility task
o The device they used for the task.
o The level of pain participants actually experienced in the task
o Demographic information (age, gender).

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